

02_Data Model & DAX

Data Model Overview

The dashboard was built using a star-schema-inspired data model centered around Orders and Order Items. Supporting dimension tables provide customer, product, seller, category, and date-level context for analysis.

Tables Used

Fact Tables

- public_orders
- public_order_items
- public_payments
- public_reviews

Dimension Tables

- public_customers
- public_products
- public_sellers
- public_category_translation
- Dim_Date

Relationships

The model uses one-to-many relationships with single-direction filtering to maintain performance and avoid ambiguity.

Relationship Structure

From Table	To Table
Orders	Customers
Orders	Payments
Orders	Reviews

From Table	To Table
Order Items	Orders
Order Items	Products
Order Items	Sellers
Products	Category Translation
Orders	Date Table

Date Dimension

A custom Date Dimension was created to support time-based analysis across all dashboard pages.

Date Table Fields

- Date
- Year
- Quarter
- Month Number
- Month Name
- Weekday
- Year Month

DAX Measures

Revenue Measures

Measure	Description
Total Revenue	Total revenue generated from payments
Total Sales	Total sales amount used across visuals
Revenue Previous Year	Previous year revenue comparison
Revenue Growth %	Year-over-year revenue growth
Revenue Per Customer	Average revenue generated per customer
Revenue Per Product	Average revenue generated per product
Top Category Revenue	Revenue generated by the highest-performing category

Measure	Description
Top Revenue State	State generating the highest revenue

Customer Measures

Measure	Description
Total Customers	Unique customer count
Monthly Customers	Customers acquired per month
Repeat Customers	Customers with multiple purchases
One Time Customers	Customers with a single purchase
Repeat Purchase Rate	Percentage of repeat customers
Orders Per Customer	Average orders placed per customer

Order Measures

Measure	Description
Total Orders	Total order count
Total Revenue Orders	Revenue-generating orders count
Category Orders	Total orders by product category

Product Measures

Measure	Description
Total Products	Total products available
Total Categories	Total product categories
Top Category Name	Highest revenue-generating category

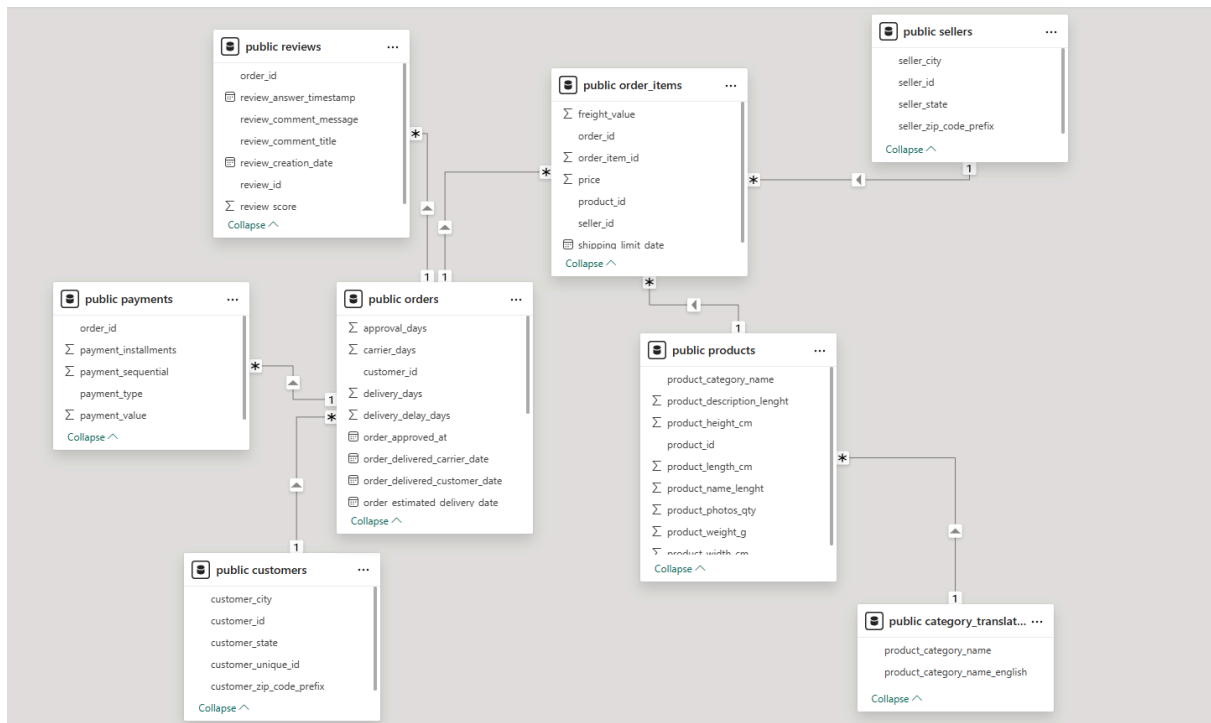
Operational Measures

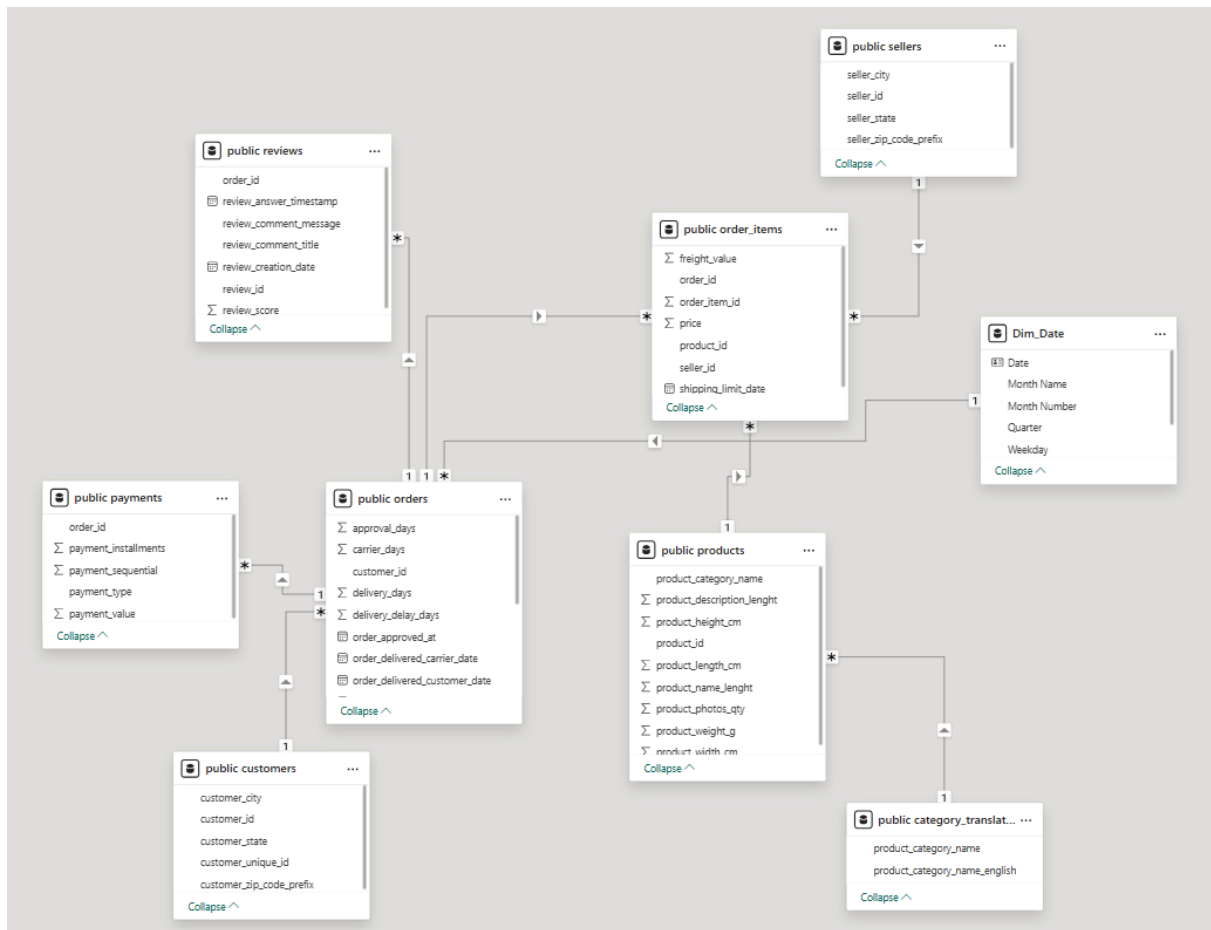
Measure	Description
Avg Delivery Days	Average delivery duration
Avg Fulfillment Days	Average fulfillment duration
Delivered Orders	Successfully delivered orders
Delayed Orders	Orders delivered after estimated date
On Time Orders	Orders delivered on or before estimated date

Measure	Description
Delay Rate %	Percentage of delayed orders
On Time Delivery %	Percentage of on-time deliveries
Delivery Success Rate	Percentage of delivered orders
Cancelled Orders	Cancelled order count
Cancellation Rate %	Percentage of cancelled orders

Customer Satisfaction Measures

Measure	Description
Total Reviews	Total review count
Average Review Score	Average customer review rating
Five Star Reviews	Total 5-star reviews
Five Star Review Rate	Percentage of 5-star reviews
Low Rating Reviews	Reviews with low ratings





Manage relationships



+ New relationship		Autodetect	Edit	Delete	Filter
From: table (column) ↑	Relationship	To: table (column)	Status		
☐ public order_items (order_id)	* — 1	public orders (order_id)	Active	...	
☐ public order_items (product_id)	* — 1	public products (product_id)	Active	...	
☐ public order_items (seller_id)	* — 1	public sellers (seller_id)	Active	...	
☐ public orders (customer_id)	* — 1	public customers (customer_id)	Active	...	
☐ public orders (order_purchase...	* — 1	Dim_Date (Date)	Active	...	
☐ public payments (order_id)	* — 1	public orders (order_id)	Active	...	
☐ public products (product_cate...	* — 1	public category_translation (pr...	Active	...	
☐ public reviews (order_id)	* — 1	public orders (order_id)	Active	...	

Date ▾	Year ▾	Month Number ▾	Month Name ▾	Quarter ▾	Year Month ▾	Weekday ▾
01-07-2017 00:00:00	2017	7	July	Q3	2017-07	Saturday
02-07-2017 00:00:00	2017	7	July	Q3	2017-07	Sunday
03-07-2017 00:00:00	2017	7	July	Q3	2017-07	Monday
04-07-2017 00:00:00	2017	7	July	Q3	2017-07	Tuesday
05-07-2017 00:00:00	2017	7	July	Q3	2017-07	Wednesday
06-07-2017 00:00:00	2017	7	July	Q3	2017-07	Thursday
07-07-2017 00:00:00	2017	7	July	Q3	2017-07	Friday
08-07-2017 00:00:00	2017	7	July	Q3	2017-07	Saturday
09-07-2017 00:00:00	2017	7	July	Q3	2017-07	Sunday
10-07-2017 00:00:00	2017	7	July	Q3	2017-07	Monday
11-07-2017 00:00:00	2017	7	July	Q3	2017-07	Tuesday
12-07-2017 00:00:00	2017	7	July	Q3	2017-07	Wednesday
13-07-2017 00:00:00	2017	7	July	Q3	2017-07	Thursday
14-07-2017 00:00:00	2017	7	July	Q3	2017-07	Friday
15-07-2017 00:00:00	2017	7	July	Q3	2017-07	Saturday
16-07-2017 00:00:00	2017	7	July	Q3	2017-07	Sunday
17-07-2017 00:00:00	2017	7	July	Q3	2017-07	Monday
18-07-2017 00:00:00	2017	7	July	Q3	2017-07	Tuesday
19-07-2017 00:00:00	2017	7	July	Q3	2017-07	Wednesday
20-07-2017 00:00:00	2017	7	July	Q3	2017-07	Thursday
21-07-2017 00:00:00	2017	7	July	Q3	2017-07	Friday
22-07-2017 00:00:00	2017	7	July	Q3	2017-07	Saturday
23-07-2017 00:00:00	2017	7	July	Q3	2017-07	Sunday
24-07-2017 00:00:00	2017	7	July	Q3	2017-07	Monday
25-07-2017 00:00:00	2017	7	July	Q3	2017-07	Tuesday
26-07-2017 00:00:00	2017	7	July	Q3	2017-07	Wednesday
27-07-2017 00:00:00	2017	7	July	Q3	2017-07	Thursday
28-07-2017 00:00:00	2017	7	July	Q3	2017-07	Friday
29-07-2017 00:00:00	2017	7	July	Q3	2017-07	Saturday
30-07-2017 00:00:00	2017	7	July	Q3	2017-07	Sunday
31-07-2017 00:00:00	2017	7	July	Q3	2017-07	Monday
01-08-2017 00:00:00	2017	8	August	Q3	2017-08	Tuesday
02-08-2017 00:00:00	2017	8	August	Q3	2017-08	Wednesday
03-08-2017 00:00:00	2017	8	August	Q3	2017-08	Thursday
04-08-2017 00:00:00	2017	8	August	Q3	2017-08	Friday
05-08-2017 00:00:00	2017	8	August	Q3	2017-08	Saturday
06-08-2017 00:00:00	2017	8	August	Q3	2017-08	Sunday
07-08-2017 00:00:00	2017	8	August	Q3	2017-08	Monday
08-08-2017 00:00:00	2017	8	August	Q3	2017-08	Tuesday